RADIANT NUCLEAR

Peak Fuel Temperature During Passive Cooldown

A 3D conjugate heat-transfer prediction for an MHTGR-350 fuel block.

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Approach

Situation

One number sets the safety case: does the fuel stay below the **1600 °C** TRISO limit?

Task

Predict the peak fuel temperature during a passive cooldown, with the context to assess its accuracy.

Action

A 3D conjugate heat-transfer model with a custom temperature-based solver and benchmark inputs.

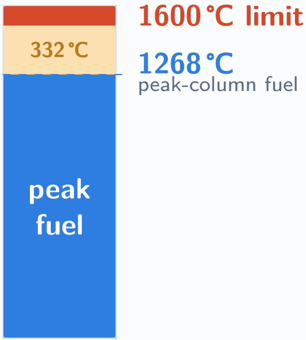
Result

Bounding peak fuel **1268 °C**, ~**332 °C** margin; the average-column baseline validates to the benchmark within 0.2%.

Situation — Safety Question

- TRISO fuel retains fission products up to about **1600 °C**.
- Keep the fuel below that limit, in normal operation and through any cooldown, and the reactor is safe by design.
- The whole thermal case reduces to one question, with a margin attached.

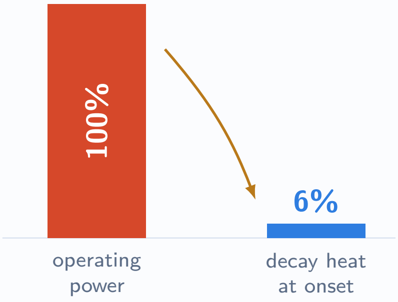
Does the fuel stay below 1600 °C during a passive cooldown, and by how much?



How close the peak gets to the 1600 °C limit.

Situation — Hottest at Full Power

- At trip, fission power collapses to about **6%** decay heat almost instantly.
- The graphite block keeps shedding stored heat to the reactor-cavity cooling system.
- So the peak is the operating value, reached at cooldown onset. The block cools monotonically after that.



Power falls about 33-fold at trip.

Task — Scope

Predict peak fuel temperatures during a passive cooldown, with the context to assess accuracy.

Scoping decisions

- One fuel block and coolant-channel column, standard HTGR practice.
- Validate in normal operation first, then apply the same model to the cooldown.
- Model the depressurised cooldown. Pressurised is core-scale, out of scope.

Deliverable
A defensible peak-fuel prediction with a stated uncertainty band.

↓

Model
A coupled steady state, then a solid-only cooldown transient.

↓

Validate
Grid convergence, energy balance, benchmark comparison.

Action — Methodology

A custom temperature-based CHT solver solves the solid directly in temperature:

$$\rho\, C_p(T)\, \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + q'''$$

The stock enthalpy formulation uses C_p inconsistently: pointwise in the diffusivity κ/C_p, but interval-averaged when converting enthalpy to temperature. These cancel only for constant C_p, so graphite's varying C_p leaves a systematic **$\sim$17 °C** error, independent of mesh.

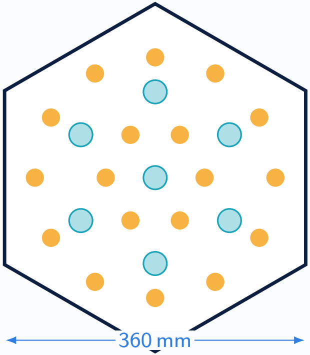
Steady baseline

Two regions, resolved helium channels coupled to the solid, marched to convergence.

Passive cooldown

Solid only. Cooling is lost and the primary is stagnant, so helium is dropped and the channels go adiabatic. Time-accurate from the steady field.

Action — Geometry & Mesh



Teal: coolant channels. Amber: fuel holes.

A real MHTGR-350 cross-section extruded to a ten-block column. Fuel compacts are a heat-source zone in one graphite region, with graded near-wall refinement.

Block across flats	360 mm
Fuel-to-coolant pitch	18.8 mm
Fuel : coolant holes	222 : 109
Column height	8.0 m
Mesh	2.64 M cells

Geometry matches the benchmark (porosity 0.186, 79.3 cm element), cross-checked against NEA/NSC/R(2017)4[1].

Action — Inputs

Operating point

Coolant, pressure	Helium, 6.39 MPa
Inlet temperature	259 °C
Inlet velocity	18.8 m/s
Power density (fuel)	24.8 MW/m^3

Decay heat

ANS-5.1 curve, about 6% at onset falling to 0.4% at 100 hours.

Materials, H-451 graphite

Density	1850 kg/m^3
Emissivity	0.85
Specific heat $C_p(T)$	benchmark fit
Conductivity $\kappa(T)$	un-irradiated

Conductivity is the un-irradiated value. Irradiation lowers κ, which runs the fuel hotter, so this input under-predicts the peak. Quantified in the accuracy assessment.

Cites: spec pp. 16–17, Tables AIV.2–AIV.3, § I.7.4; full trace in `benchmark/README.md`.

Action — Boundary Conditions

Surface	Steady	Passive cooldown
Coolant channels	coupled to solid	adiabatic
Outer wall	adiabatic	radiation to the RCCS sink
Axial ends	adiabatic	adiabatic

Placing the RCCS sink on the block itself zeroes the core radial resistance. It really belongs at the vessel boundary of a full-core model. This is the main reason the block shows no delayed peak.

The wall condition is effectively radiation-only ($\varepsilon = 0.85$, sink at 303 K). Adiabatic channels trap heat, which is conservative for the peak.

Action — Assumptions

- Single block and column, not the annular core.
- Un-irradiated conductivity.
- ANS-5.1 decay heat.
- Uniform power in the axial direction.
- Depressurised cooldown only.

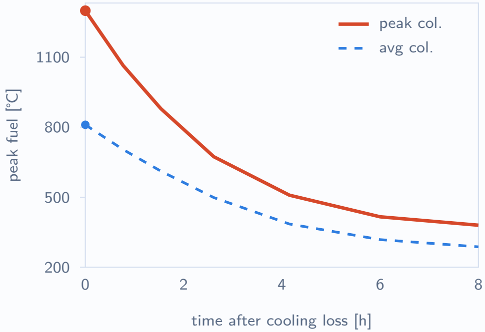
Why these hold

A single symmetric block is the standard unit for prismatic-HTGR analysis: representative physics on a mesh clean enough to validate.

Un-irradiated κ was the only value in the distributed data; its low bias is quantified in the accuracy assessment.

The depressurised case is the bounding conduction cooldown, so it needs no active-fluid model.

Result — Prediction



Both columns cool monotonically from their steady peak. No delayed peak at block scale.

1268 °C
peak-column fuel

∼332 °C
margin to the limit

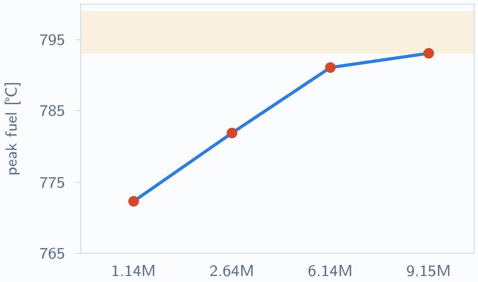
The peak-power column (×**1.85**) is the bounding safety number: within 1% of the benchmark core max (1282 °C[3]). The average column (**793 °C**) validates the benchmark outlet. Both cool from their steady peak.

Result — Validation

Four meshes spanning eight-fold in cell count, average-power column.

Mesh	Peak fuel	Outlet
1.14 M	772.3	666.7
2.64 M	781.9	676.3
6.14 M	791.1	685.5
9.15 M	**793.1**	**688.2**

Converged near second order (GCI about 1%). Coolant outlet **688 °C** against the benchmark **687 °C**[1], a 0.2% agreement.



Peak and outlet converge together as the coolant mixing resolves.

Result — Verification

$$\Delta\int_V \rho h\, dV \;=\; \int_0^t\!\int_V q'''\, dV\, dt \;-\; \int_0^t\!\oint_A \kappa\, \nabla T\cdot\hat{n}\, dA\, dt$$

stored-enthalpy change = decay heat generated − heat radiated to the RCCS

- Average column closes to <**0.25%** ($\Delta U = -584\,$MJ), and **0.80%** on the finest mesh.
- Peak column closes to **1.84%**, a C_p-discretisation effect that leaves the reported peak unaffected.
- **Time-step independent**: peak fuel 778.0 °C at both time steps.

Cooldown time constant

$$\tau = \frac{\rho\, c\, V}{h_{\text{rad}}\, A} \approx 128\ \text{min}$$

CFD cooldown: **121 min**, within 5%.

Result — Accuracy

Source	Effect	Next step
No core radial path (block, not core)	under-predicts	effective-core model
Adiabatic channels	over-predicts	none needed
Un-irradiated conductivity	under-predicts	use irradiated κ

The average column under-predicts. Carry the peak-power column (**1268 °C**, $\approx$ benchmark core max, margin $\sim$332 °C) as the bounding block-scale estimate. The block does not resolve the whole-core delayed peak, the next-fidelity step.

Result — Next Steps

What is established

- Steady baseline validated to the benchmark outlet within 0.2%, and grid-converged.
- Transient integrator verified: energy-conservative and time-step independent.
- Every input, boundary condition, and assumption stated and traceable.

What it is not

- It resolves one block, not the annular core, so it does not produce the whole-core delayed peak.

Next step

Restore the radial heat path with an effective-core model, moving the RCCS sink to the vessel boundary. That model can be compared directly against the benchmark full-core peak: 1391 K[2] at 50 h, or 1237 K at 74 h for the ring model.

This verified block model and its validated baseline are the foundation it builds on.

Summary

Peak fuel temperature during passive cooldown

	Peak fuel	Margin to 1600 °C
Peak-power column (×1.85, hottest)	**1268 °C**	∼332 °C
Average-power column	793 °C	∼807 °C

Steady coolant outlet 688 °C (benchmark 687 °C); peak at cooldown onset, then monotonic cooldown.

Operating conditions

Coolant, pressure	Helium, 6.39 MPa
Core inlet	259 °C, 18.8 m/s
Power density (avg)	24.8 MW/m^3
Power density (peak)	46.0 MW/m^3
Decay heat	ANS-5.1, ∼6%

Geometry

Block	MHTGR-350 prismatic
Across flats	360 mm
Hole pitch	18.8 mm
Fuel : coolant holes	222 : 109
Column	8.0 m, 10 blocks

Sources

1 OECD/NEA MHTGR-350 benchmark, NEA/NSC/R(2017)4. Geometry, operating point, materials (Tables AIV.2–AIV.3), decay heat (§ I.7.4), and the steady coolant outlet (687 °C).

2 Strydom et al., INL. MHTGR-350 DCC/PCC transient reference results: the full-core delayed peak, 1391 K at 50 h (block) and 1237 K at 74 h (ring).

3 MHTGR-350 steady multiphysics results: the core-maximum fuel temperature (≈1282 °C), a secondary reference band.

Benchmark values and page cites are catalogued in `benchmark/README.md`.